

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		vertical part of titration curve is between 6.5 (allow ± 0.5) and 11 (allow ± 1) / vertical part lies in alkaline region (1) select an indicator whose colour change is complete in the vertical region (1)	2			2		2
		(iv)	I	$\text{HF} \rightleftharpoons \text{H}^+ + \text{F}^-$ / reversible reaction (or equilibrium) (1) addition of a small amount of acid reacts with fluoride ions / shifts equilibrium to left (1) addition of a small amount of base removes hydrogen ions and these are replaced by HF dissociating / shifting equilibrium to the right / hydroxide ions react with HF molecules (1)	3			3		
			II	concentration of acid equal to concentration of salt (1) $\text{pH} = -\log [7.2 \times 10^{-4}] = 3.14$ (1)		2		2	1	
				Question 8 total	7	7	0	14	4	4